

Investigate the impact of Confounding Medical Interventions (CMI) in electronic health records (EHR) data

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Background & Motivation

Electronic health records (EHRs) data are widely used for clinical research. The abundance of available data, the low cost of obtaining those observational data, and its real-time nature have given EHR enormous potential to develop powerful predictive models. Researchers rely on EHR data extensively to predict the risk of clinical outcomes. As clinical prediction models (CPMs) find their way into clinical practice, it is important to develop systems and procedures to monitor their safety and effectiveness. The Food and Drug Administration (FDA) has developed draft guidance for how such tools should be considered. Within Duke University Health System, a governance framework has been developed for evaluating and monitoring the performance of CPMs to ensure their adequate performance.

After implementing a CPM, medical professionals will use these models to inform clinical decision-making and allocation of interventions. Ideally, the intervention will prevent the outcome that the CPM is predicting. This leads to an important challenge: how do we evaluate CPMs in the presence of interventions? If the CPM is accurate, and the intervention is effective, those who are truly at risk of the adverse outcome will not actually have the event. To evaluate the CPM appropriately, we are interested in what the causal inference refers to as the counter-factual: the outcome that would have occurred without the intervention being deployed. Unfortunately we do not observe this counter-factual.

Current research on EHR data has investigated several sources of biases such as selection bias informed presence bias, and confounding, and proposed approaches to control these biases. However, little work has considered this potential bias which has been labeled: Confounding by Medical Interventions (CMIs). The little literature on this topic has focused on how we can learn new CPMs in the presence of CMI. The suggestions to mitigate this bias include using data with no CMI for model training or treating all observations with CMI as unhealthy (based on the trust in medical professional's judgement). Although the researchers were able to improve the classification accuracy using these methods, ignoring data with CMI during training reduced the generalizability of the model to validation data, and assigning all data with CMI with negative labels did not perform well enough. As such, the literature is not well developed. This project seeks to study how CMI can impact our ability to evaluate CPMs and strategies we can use to address these challenges. We first propose possible causal models. Next, we use simulation to both illustrate the problem and explore potential solutions. Then, we will evaluate these approaches in real-world healthcare data.

Exploratory analysis

During the initial stage of investigation, we proposed 2 possible causal models.

The basic variables in the causal models are initial risk, intervention (1 = intervention, 0 = no intervention), prevalence of disease, and observed outcome (1 = sick, 0 = healthy). In a simulated scenario, a patient with initial risk higher than the intervention cutoff (CMI) will receive intervention treatment. The effectiveness of treatment is decided by $b_{intervention}$. In general, a higher $b_{intervention}$ means lower effectiveness. We simulate three levels of disease prevalence: high (~50% population have the illness without intervention), moderate (~30% population have the illness without intervention), and low (~18% population have the illness without

intervention). A highly prevalent disease includes oral disease, according to a November 2022 WHO brief. A moderately prevalent disease includes Hepatitis B, and a low-prevalent disease includes kidney disease. What differentiates the 2 models is the mechanism to generate the final observed outcome with intervention. The first model decides a new risk (risk after intervention) for patients who should have had outcome due to higher initial risk and received intervention treatment. According to the new risk, the final observed outcome is generated probabilistically. The second model doesn't consider risk after intervention. Instead, depending on the intervention effectiveness factor $b_{intervention}$, $(1-b_{intervention})$ % of the population will be assigned to have no outcome (healthy), which means their intervention is successful.

In real world, only the initial risk and observed outcome are known. With the confounding intervention variable, we ideally want to achieve a prediction power as good as predicting outcome directly from the known initial risk. The metric values (AUC, MSE, deviance, calibration curve slope) are averaged over 100 simulations.

Causal diagram 1

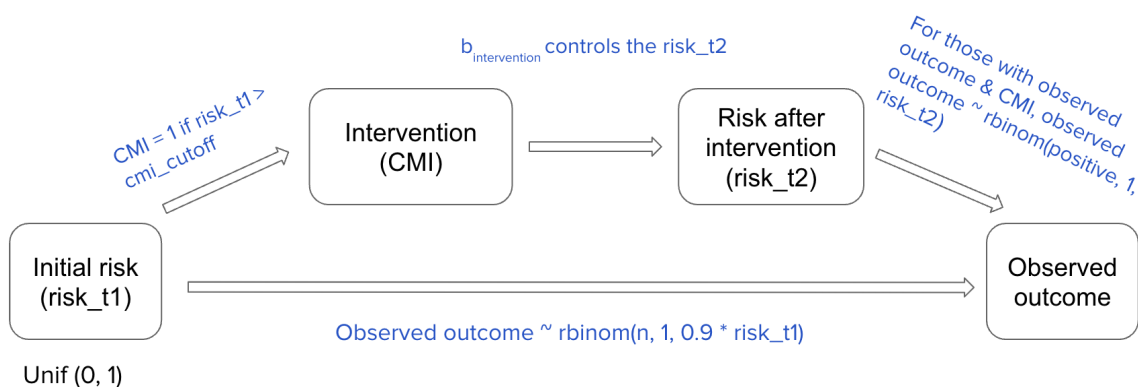


Figure 1: Causal diagram 1

```
## [1] 0.172
```

```
## Area under the curve: 0.7058
```

Vary $b_{intervention}$ (effectiveness of treatment)

```
## [[1]]
```

```
## [1] 23.08896 20.25403 20.49730 19.97940 19.13856 18.66202 18.67977 17.19305
```

```
## [9] 16.66029 15.75249
```

```
##
```

```
## [[2]]
```

```
## [1] 15.645805 14.161590 13.505084 12.648312 12.045015 11.405450 10.212787
```

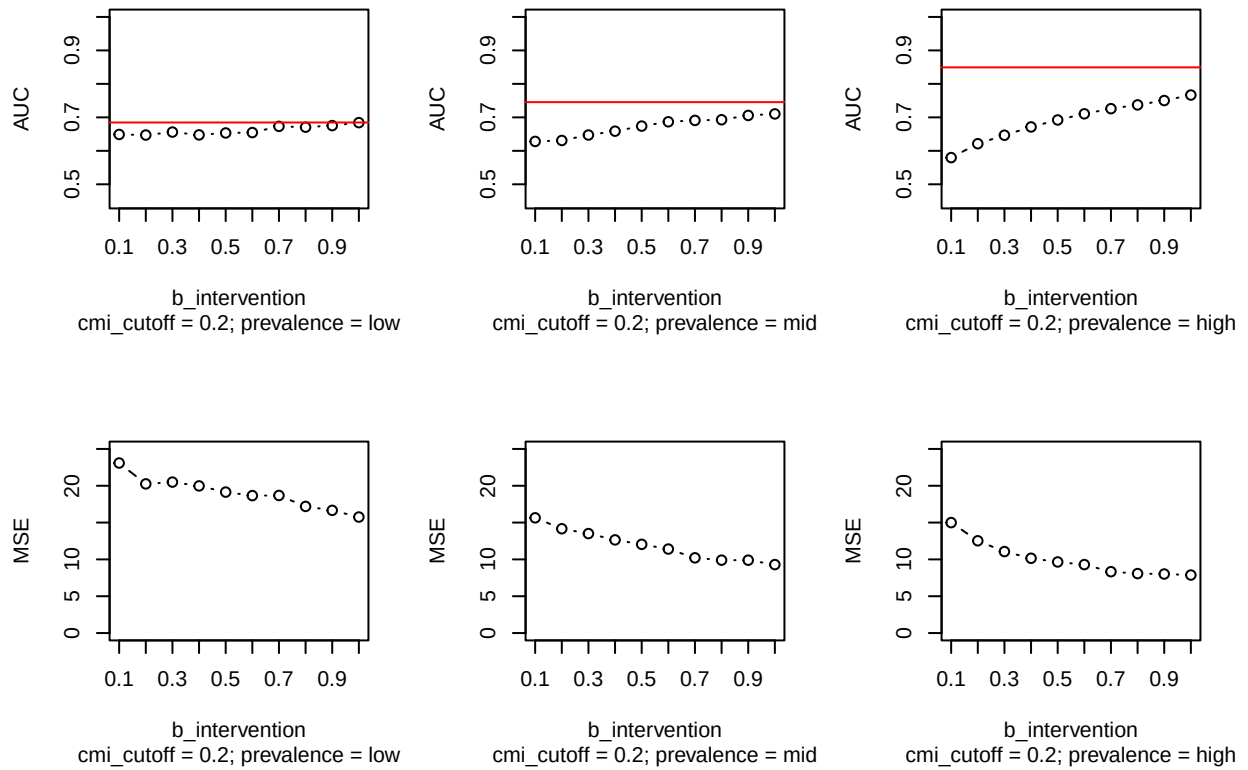
```
## [8] 9.898509 9.894571 9.295035
```

```
##
```

```
## [[3]]
```

```
## [1] 14.996041 12.529303 11.065033 10.154014 9.652066 9.294096 8.330388
```

```
## [8] 8.075072 8.013839 7.874956
```

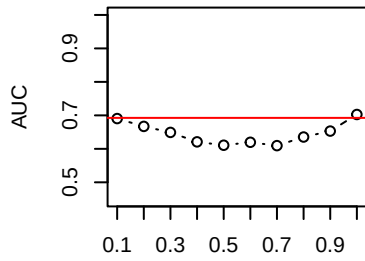


The red line denotes the prediction accuracy when fitting the observed outcome on the initial risk. As the intervention effectiveness decreases, prediction accuracy increases. A strong treatment would make a poor prediction on a patient's final outcome. As a disease becomes less prevalent, the overall prediction accuracy decreases, but the treatment effect has less effect on prediction accuracy. For a highly prevalent disease, when the intervention effectiveness is the strongest, the AUC is around 0.5, meaning it's no better than flipping a coin.

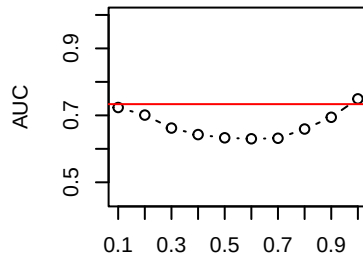
The mean squared error (MSE) is calculated as $\text{mean}(\text{fit}\$residuals^2)$ (taking the difference between fitted and actual response values) where fit is a glm function. It decreases as prediction accuracy increases.

Vary cmi_cutoff (criteria of giving out treatment)

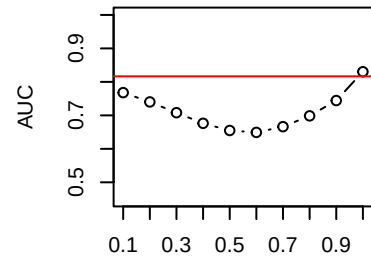
```
## [[1]]
## [1] 17.977644 15.995133 16.054056 13.202326 11.397786 10.708255 9.718422
## [8] 9.011205 8.648557 8.981789
##
## [[2]]
## [1] 11.591603 10.212894 8.649956 7.256767 6.337453 5.990785 5.557892
## [8] 5.281792 5.235374 5.747313
##
## [[3]]
## [1] 9.182648 8.406228 6.727133 5.522765 4.969995 4.679253 4.531430 4.564572
## [9] 5.052844 6.359651
```



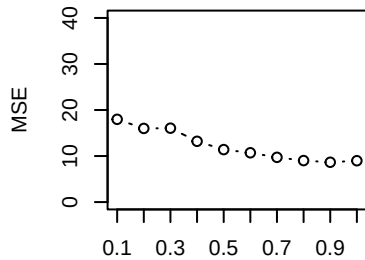
cmi_cutoff
b_intervention = 0.8; prevalence = low



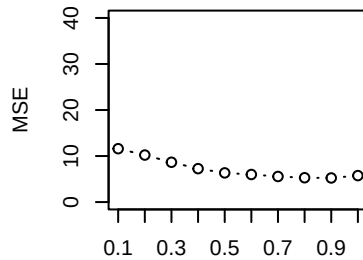
cmi_cutoff
b_intervention = 0.8; prevalence = mid



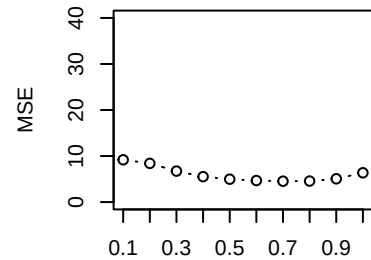
cmi_cutoff
b_intervention = 0.8; prevalence = high



cmi_cutoff
b_intervention = 0.8; prevalence = low



cmi_cutoff
b_intervention = 0.8; prevalence = mid



cmi_cutoff
b_intervention = 0.8; prevalence = high

MSE show a similar trend when we vary the cmi_cutoff. Notice that the higher the cutoff, less people are getting the intervention.

Varying CMI_cutoff shows a different trend of prediction accuracy from varying intervention effectiveness. To see a broader picture, we want to see how CMI_cutoff and intervention effectiveness interact to affect the metrics.

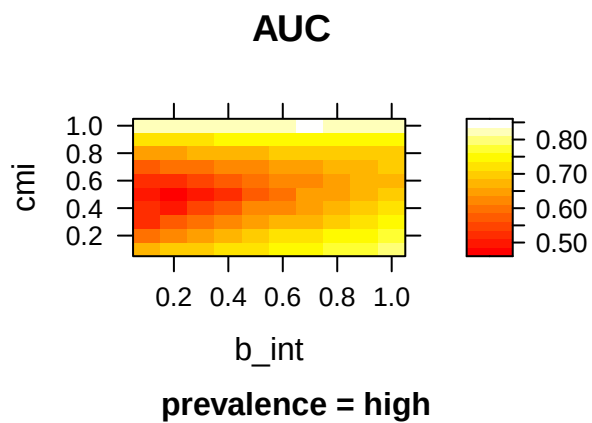
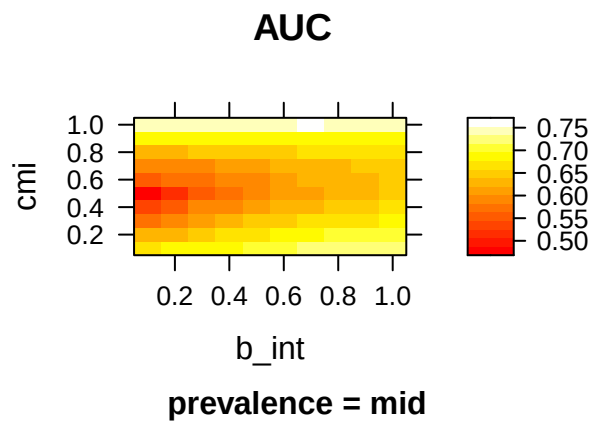
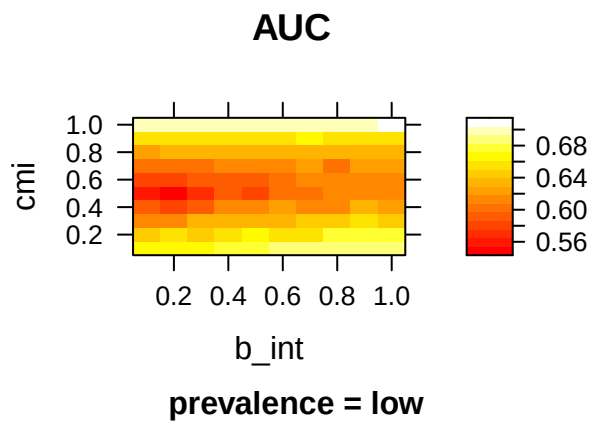
Varying b_int and cmi_cutoff

```
## [[1]]
##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]
## [1,] 1.0974759 1.2647127 1.0667921 1.6620046 0.6068377 1.2035958 1.1879634
## [2,] 1.1399970 1.1721788 1.4225685 0.5664675 -0.3165409 1.7109683 1.1607994
## [3,] 1.0087681 1.1415940 1.3417690 1.0255246 0.2125518 3.5599965 1.0508586
## [4,] 1.0209604 1.2345581 0.6449650 1.4968154 1.6503697 0.9417691 1.0312863
## [5,] 0.9148139 1.1805590 1.2266997 1.2468584 1.1350578 0.9224123 1.9385641
## [6,] 1.0804077 1.0547799 1.1483852 1.1154136 1.1159530 1.4030549 1.0732366
## [7,] 0.9625322 1.1204221 1.0985095 1.1759328 1.2007774 1.5174917 1.0157206
## [8,] 1.0899964 1.1339069 1.0499324 1.1798136 1.1792587 1.4148865 0.9205024
## [9,] 1.0127185 1.0533908 1.0804070 1.2505222 0.8301656 1.1163567 1.0025132
## [10,] 0.9460494 0.9790709 0.9654339 1.0046892 1.1419763 1.0231691 0.9571342
##           [,8]      [,9]     [,10]
## [1,] 0.9978863 0.9279323 1.0030135
## [2,] 0.9318816 0.9748692 0.9994114
## [3,] 0.9857016 0.9716495 0.9643148
## [4,] 1.0239993 0.9530648 0.9808372
## [5,] 1.0237636 1.1316753 0.9593340
## [6,] 1.0042989 0.9581222 0.9419393
## [7,] 0.9126477 1.0213136 1.0084687
## [8,] 0.9721040 0.9125471 0.9780294
```

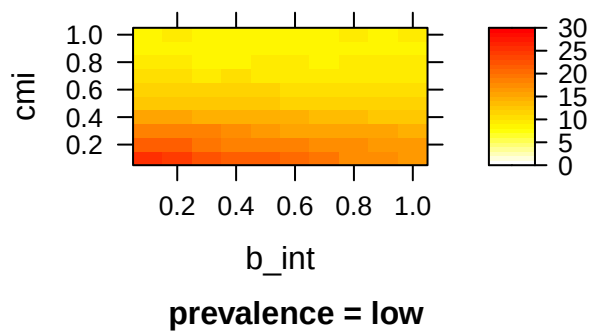
```

## [9,] 1.0046785 0.9819344 0.9570324
## [10,] 1.0412604 0.9214042 1.0069679
##
## [[2]]
##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]
## [1,] 1.1755416 1.2844206 2.050091 0.129136 1.0863430 1.5396595 1.0476241
## [2,] 1.0953240 1.1893362 1.175804 1.383054 -0.1127267 1.1971737 1.1599517
## [3,] 1.0456765 1.1800368 1.401742 1.666569 1.2592388 -1.1302293 0.9278280
## [4,] 1.0203663 1.1741694 1.149415 1.104622 -0.5906191 1.1030042 1.0307134
## [5,] 0.9968458 1.0661480 1.151157 1.199844 1.1595836 1.1474209 0.9927551
## [6,] 1.0114513 1.0393894 1.169810 1.069680 1.0594038 1.0701581 0.9210752
## [7,] 1.0362302 1.0387071 1.089212 1.008929 1.1174461 0.9563726 1.0627397
## [8,] 1.0006870 1.0108281 1.033687 1.060844 0.9720262 1.0055299 0.8969308
## [9,] 0.9959034 1.0373963 1.032905 1.019110 0.9647678 0.9799221 0.9753578
## [10,] 0.9872233 0.9935796 1.032700 1.027094 0.9675728 0.9807394 1.0046977
##           [,8]      [,9]      [,10]
## [1,] 0.9828520 0.9527008 0.9795338
## [2,] 0.9703468 0.9780914 0.9813213
## [3,] 0.9822473 0.9585838 0.9797316
## [4,] 0.9316580 0.9334651 0.9990101
## [5,] 0.9096186 0.9618051 0.9982619
## [6,] 0.9529077 0.9367351 0.9930632
## [7,] 0.9279167 0.9650593 1.0162103
## [8,] 0.9783956 0.9838751 0.9976597
## [9,] 1.0162177 0.9479427 0.9817762
## [10,] 0.9463998 1.0030013 0.9724425
##
## [[3]]
##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]
## [1,] 1.2498837 1.3791111 0.812338 -5.1630012 1.7253473 0.9557473 1.0911258
## [2,] 1.0395025 1.344972 1.440091 -0.4706524 26.8862234 -2.6212570 0.9687649
## [3,] 1.0411527 1.154816 1.238101 -2.3437842 1.2877059 0.9800746 1.0164834
## [4,] 0.9773406 1.090518 1.128813 1.0930457 1.1140340 1.0712672 0.9782777
## [5,] 0.9966584 1.007043 1.116251 1.0807220 0.9891114 1.0782213 0.9826688
## [6,] 1.0069627 1.016464 1.041661 1.0195388 1.0045226 1.0169889 0.9814193
## [7,] 1.0056194 1.041379 1.025942 1.0114923 0.9940594 0.9474737 0.9521856
## [8,] 1.0031378 1.018505 1.037145 1.0081833 0.9980909 0.9752685 0.9447942
## [9,] 0.9895918 1.024485 1.049244 0.9848526 1.0121605 1.0042189 0.9617050
## [10,] 1.0097352 1.034639 1.016358 1.0085997 0.9836207 0.9665536 0.9923487
##           [,8]      [,9]      [,10]
## [1,] 0.9740079 1.0097407 0.9754368
## [2,] 1.0094719 0.9963573 0.9982192
## [3,] 0.9790389 0.9999461 0.9912247
## [4,] 0.9780567 1.0047914 0.9798945
## [5,] 0.9757878 0.9945170 0.9863264
## [6,] 1.0306645 1.0003210 0.9928984
## [7,] 0.9887086 1.0339635 0.9995081
## [8,] 1.0026568 0.9912975 0.9873119
## [9,] 0.9799345 1.0252616 0.9913652
## [10,] 0.9576530 1.0083757 0.9879842

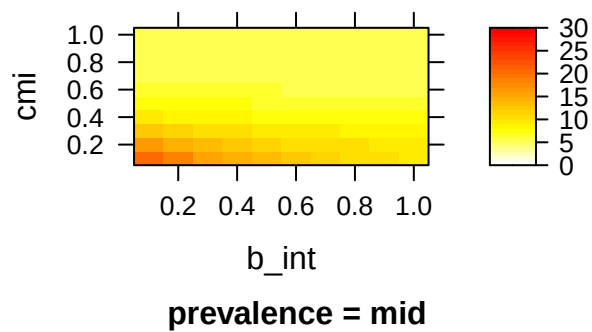
```



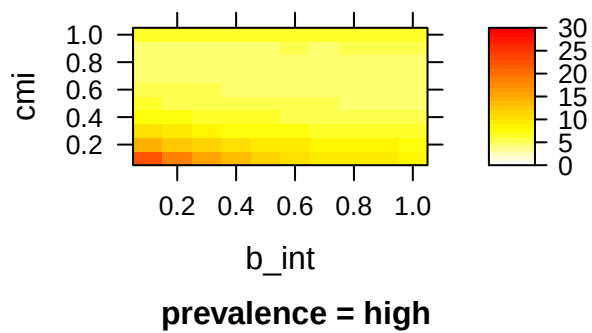
Mean Squared Error



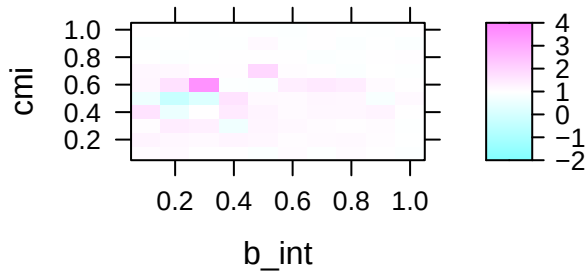
Mean Squared Error



Mean Squared Error

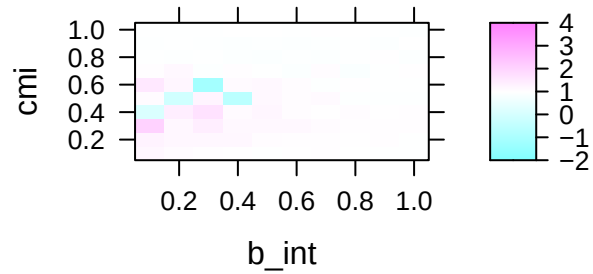


Calibration curve slope



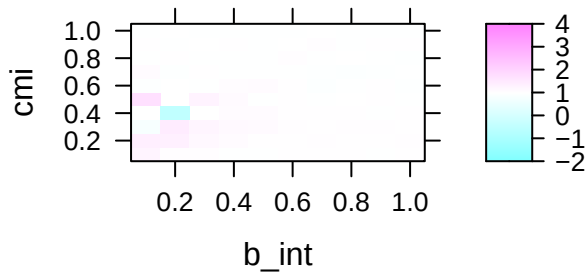
prevalence = low

Calibration curve slope



prevalence = mid

Calibration curve slope



prevalence = high

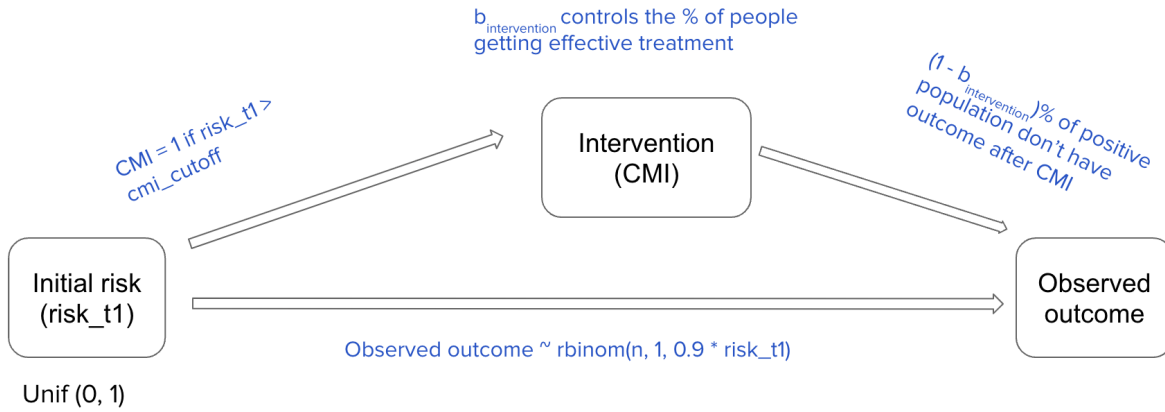
Moving from bottom to top, if close to half of population receives treatment, the prediction performance is the worst. However, if almost everyone or no one receives treatment, we see better performance. Moving from left to right, as intervention effectiveness decreases, AUC increases. This trend is more clear in moderate and high prevalent scenario.

MSE shows that as intervention effectiveness decreases or as less people receives treatment, there is less error in prediction. MSE is overall higher for lower-prevalence diseases.

Causal diagram 2

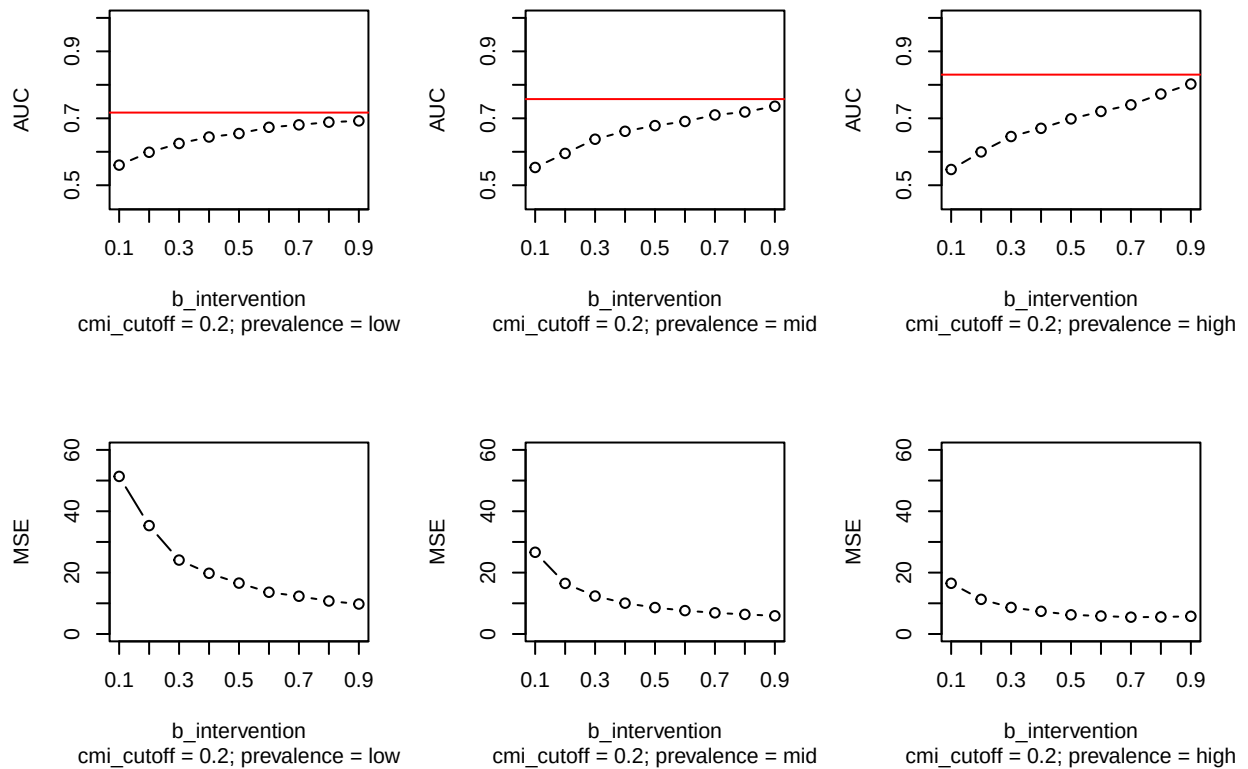
vary b_intervention

```
## [[1]]
## [1] 51.354870 35.323392 24.088914 19.781156 16.583544 13.601456 12.308728
## [8] 10.744120 9.764164
##
## [[2]]
## [1] 26.628982 16.483287 12.364906 10.045031 8.606806 7.621689 6.892779
## [8] 6.384072 5.927106
##
## [[3]]
## [1] 16.520102 11.286958 8.644583 7.384969 6.256682 5.853996 5.483436
## [8] 5.545641 5.758304
```



*Positive population: those with observed outcome & CMI

Figure 2: Causal diagram 2



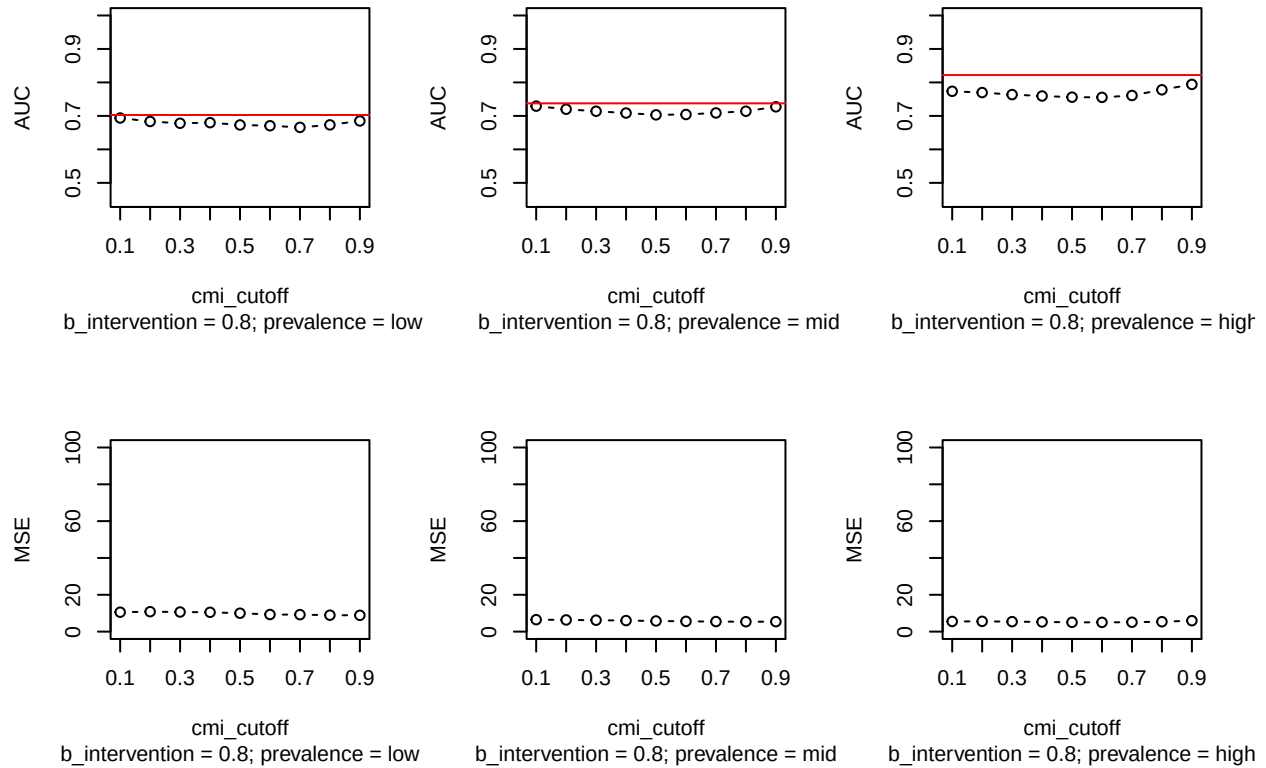
MSE and AUC shows similar trend as the first causal diagram. However, in the second causal model, the case when intervention almost has no effect has an AUC very close to the AUC under the direct relationship between health outcome and initial risk.

vary cmi_cutoff

```
## [[1]]
## [1] 10.537485 10.815052 10.642021 10.486916 9.990921 9.278383 9.215241
## [8] 8.923568 8.853030
```



```
##
## [[2]]
## [1] 6.484965 6.334776 6.197875 5.991503 5.774307 5.569014 5.454496 5.374978
## [9] 5.403168
##
## [[3]]
## [1] 5.502190 5.565944 5.403506 5.254297 5.042735 5.026541 5.093181 5.365917
## [9] 5.941068
```



The effect of cmi_cutoff on prediction accuracy is relatively consistent across different values as shown in both AUC and MSE plots. In general, high prevalence of disease leads to higher prediction accuracy no matter what value cmi_cutoff takes. Prediction accuracy large depends on the disease prevalence and effectiveness of treatment.

Vary both b_intervention and cmi_cutoff

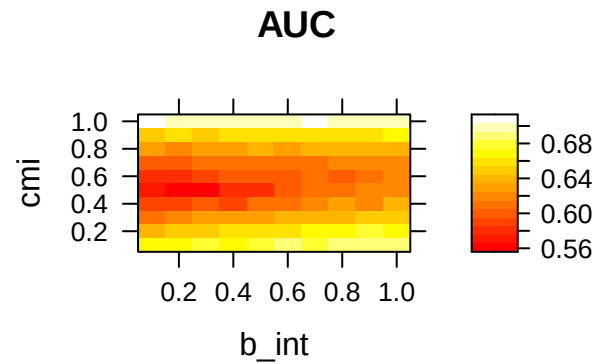
```
## [[1]]
##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]
## [1,] 1.0941634 1.324748 0.9582097 1.3011024 0.8491092 0.4829402 1.0414031
## [2,] 0.7754422 1.153025 1.3291925 1.4366981 -0.9645563 1.0943253 1.1280236
## [3,] 1.1378810 1.052609 1.2017254 1.1196528 1.5804283 0.2164498 1.1861561
## [4,] 1.0557581 1.111959 1.3659811 1.1827437 1.3499000 1.2357921 -3.3855351
## [5,] 0.9848275 1.105618 1.1644355 1.2010231 1.1397132 1.2816314 1.0132422
## [6,] 1.0494237 1.068391 1.0918187 1.1577813 2.3201413 1.1844355 1.0549082
## [7,] 0.9626409 1.203316 1.1525851 0.8635708 1.0572296 1.0608280 0.9158868
## [8,] 1.0257553 1.069895 1.1539325 1.0738644 1.2932435 0.9577129 1.0484241
## [9,] 1.0171007 1.082868 1.1178049 1.1113293 1.1258499 1.1090939 1.0521187
## [10,] 1.0179057 1.107264 1.0238806 1.1825237 1.0737926 1.1882818 0.9499481
##           [,8]      [,9]     [,10]
## [1,] 1.0544679 0.9364667 0.9784640
```

```

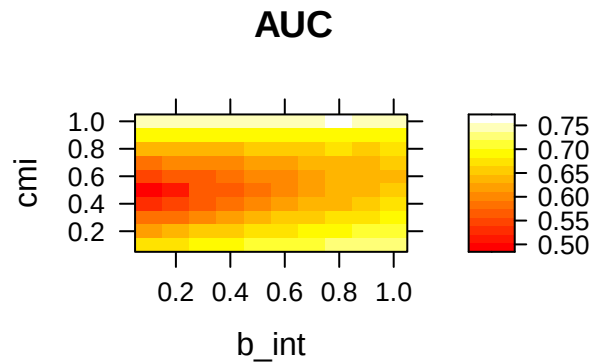
## [2,] 0.8867002 0.9618711 0.9485573
## [3,] 0.9688265 0.9578320 0.9549041
## [4,] 0.9743555 0.9556093 1.0276844
## [5,] 1.0065988 0.9505279 1.0108490
## [6,] 0.9099561 0.9435807 1.0081071
## [7,] 0.9832695 0.9049048 1.0302179
## [8,] 1.0113629 0.9351522 0.9313899
## [9,] 0.9855640 0.9907053 0.9969750
## [10,] 0.9199678 0.9768049 0.9881745
##
## [[2]]
##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]
## [1,] 1.3529506 1.0924937 1.5457239 1.1010835 2.7329135 0.8257363 0.9955407
## [2,] 0.9844712 1.1662882 1.1621084 -0.0735238 3.6597708 0.9387662 1.2771050
## [3,] 1.0338122 1.1451176 0.8034131 1.1983772 1.0390461 1.3397269 1.0815454
## [4,] 1.0072623 1.0860426 1.0987522 1.1631540 -2.8536688 0.9675996 1.0373763
## [5,] 1.0555628 1.0946237 1.1953470 1.0727624 1.0236063 0.8071395 1.0017784
## [6,] 0.9758511 1.0095340 1.1928670 1.0282825 1.0580323 0.9073882 0.9381833
## [7,] 0.9686610 1.0041718 1.0377187 1.0432251 1.2090375 1.0452267 0.9322809
## [8,] 1.0009628 0.9754375 1.0987905 1.0457610 1.0575059 1.0182818 0.9535902
## [9,] 0.9419303 1.0265323 1.0767231 1.0420631 1.0229022 0.9780647 0.9338998
## [10,] 0.9795777 1.0048804 1.0318424 1.0372629 0.9751495 0.9865372 0.9528210
##           [,8]      [,9]      [,10]
## [1,] 0.9515787 0.9423175 0.9710858
## [2,] 0.9630955 0.9858678 0.9954855
## [3,] 0.9287116 0.9594208 1.0114961
## [4,] 0.9617515 0.9159762 0.9992663
## [5,] 0.9686245 0.9895267 1.0198654
## [6,] 0.9272688 0.9481018 0.9774003
## [7,] 0.9509778 0.9584449 0.9835255
## [8,] 0.9796748 0.9494866 1.0098360
## [9,] 0.9714118 0.9369179 0.9807784
## [10,] 0.9609500 0.9817985 0.9948377
##
## [[3]]
##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]
## [1,] 1.0562609 -1.373678 2.948028 1.5418479 0.8865305 0.3871468 0.7792449
## [2,] 1.0486349 1.395269 1.576258 -1.5768618 -0.4304146 1.3580542 1.0096538
## [3,] 1.0801387 1.103265 1.304466 0.6863301 1.1092376 0.7860772 1.0315064
## [4,] 1.0028430 1.120216 1.134191 1.1422514 1.5721973 0.9169387 0.9632404
## [5,] 1.0163014 1.086714 1.104320 1.1049580 1.0939768 1.0059236 0.9706964
## [6,] 0.9877198 1.023665 1.092878 1.0623669 1.0118082 0.9923701 0.9619941
## [7,] 0.9995967 1.040020 1.051308 1.0424893 0.9907573 0.9724311 0.9378869
## [8,] 0.9880021 1.019476 1.042330 1.0240402 1.0262582 0.9935236 0.9720746
## [9,] 1.0104175 1.040790 1.044944 1.0160389 1.0130088 0.9707578 0.9858591
## [10,] 0.9872934 1.020812 1.046514 1.0225959 1.0100429 0.9839509 0.9468256
##           [,8]      [,9]      [,10]
## [1,] 0.9836591 0.9860917 0.9801830
## [2,] 0.9883748 1.0111668 0.9964978
## [3,] 0.9957058 1.0051000 0.9792681
## [4,] 0.9685539 0.9960419 0.9952156
## [5,] 1.0019249 0.9871383 0.9813348
## [6,] 1.0161770 1.0041712 1.0023461
## [7,] 0.9825263 0.9974436 0.9877215

```

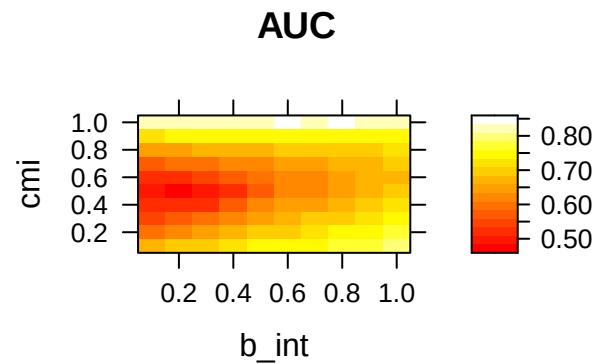
```
## [8,] 1.0072398 1.0151872 0.9848901
## [9,] 1.0226571 0.9886249 0.9841442
## [10,] 1.0001868 0.9982220 0.9876429
```



prevalence = low

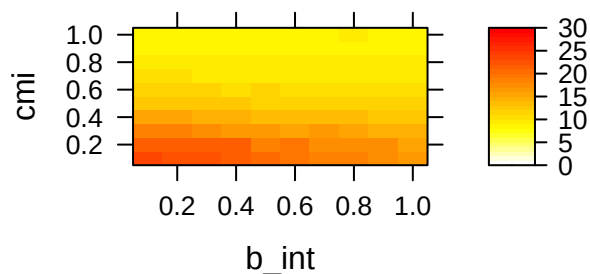


prevalence = mid



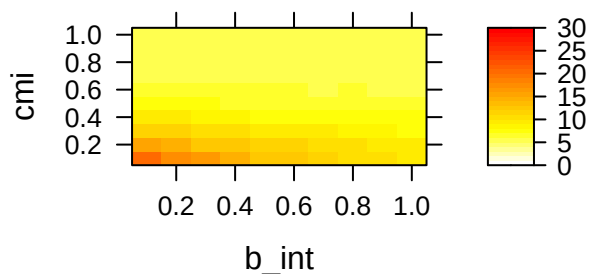
prevalence = high

Mean Squared Error



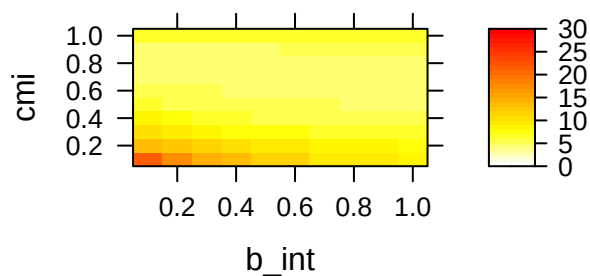
prevalence = low

Mean Squared Error



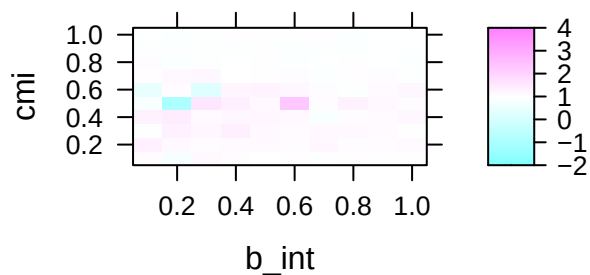
prevalence = mid

Mean Squared Error



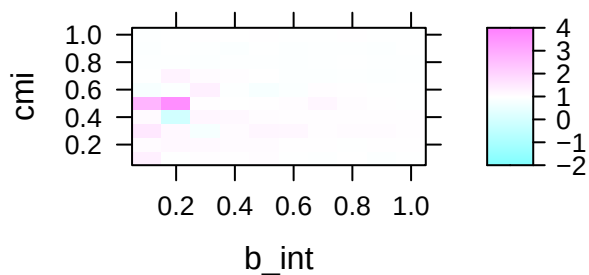
prevalence = high

Calibration curve slope



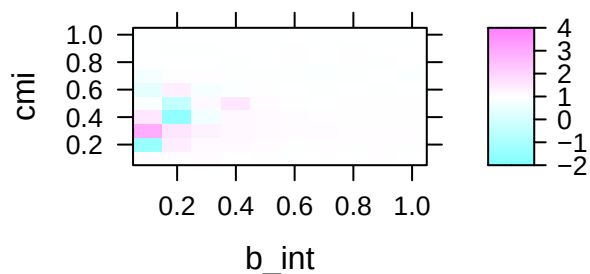
prevalence = low

Calibration curve slope



prevalence = mid

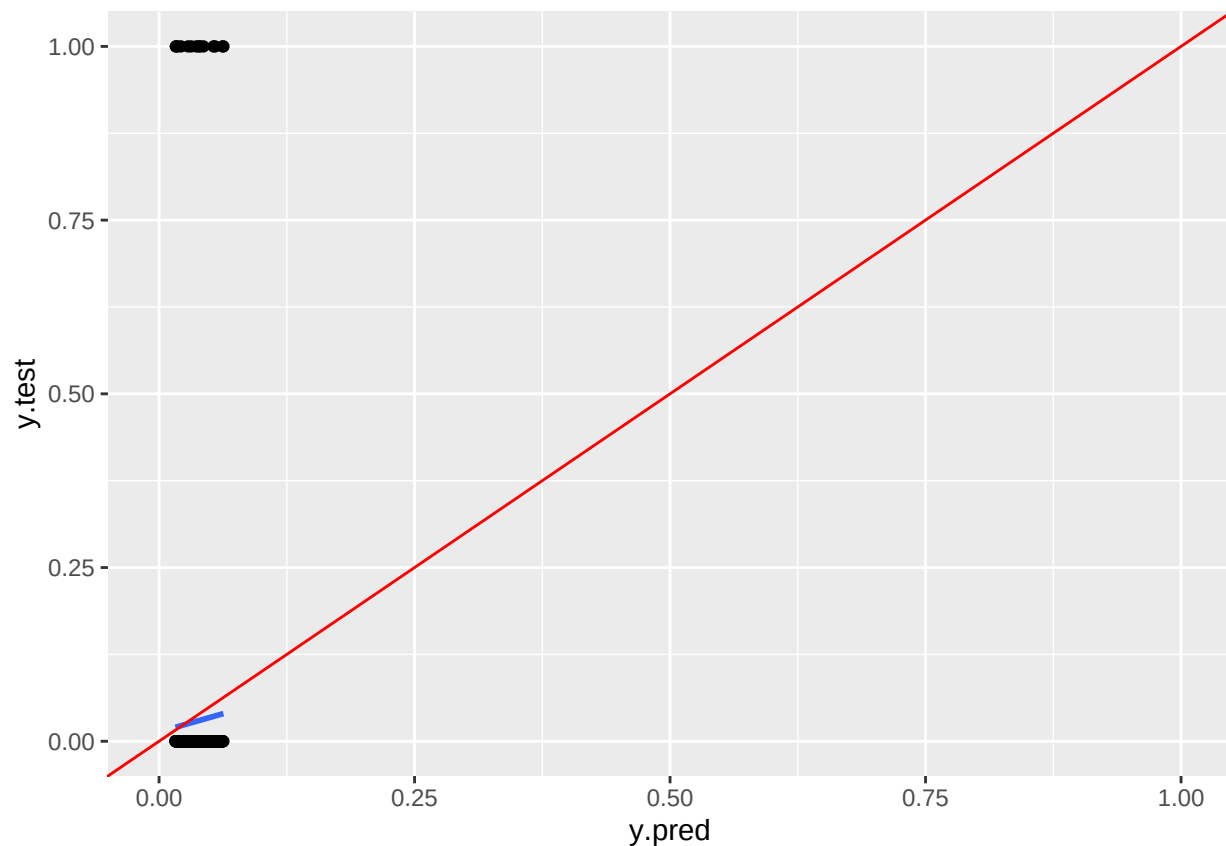
Calibration curve slope



prevalence = high

Trends are similar to the first causal model.

Is this the correct way to find calibration slope? Some calibration curve slope have negative slope (see below). The way I find calibration slope is $\text{calibration fit} = \text{lm}(y.\text{test} \sim y.\text{pred}, \text{data} = \text{data_test})$ slope = $\text{calibration fit}\$coefficients[2]$



Research question

From the exploratory analysis, we found that as the effectiveness of intervention treatment increase or the prevalence of disease decrease, the predictive performance of our glm model has decreased and MSE has increased. When less people receives treatment, there is less error in prediction. These discoveries circle back to our research question — evaluating CPMs in the presence of intervention. CPM prefers (produce better predictive performance and lower error) high-prevalence disease, poor treatment, and small coverage of intervention.

What to do next - with the goal being improving predictive power * explore other classification model other than glm, such as random forest or boosting * As treatment become stronger, there end up to be fewer positive samples. Maybe we can try upweighing positive samples as we increase the power of treatment * decompose total effect into direct and indirect effects * <https://cran.r-project.org/web/packages/foreach/vignettes/foreach.html> optimize time complexity

foreach library to optimize time complexity